Pricing A/B Test Report

Practice answers, cleaned data summary, statistical results, and recommendation

Executive Conclusion

The $59 price reduced conversion but increased revenue per visitor. Conversion fell from 1.99% to 1.56%, while revenue per visitor rose from $0.776 to $0.918. The revenue lift was $0.142 per visitor (18.25%), p-value 1.10e-08.

Recommendation: treat the higher price as promising, but do not ship solely on the raw result. Rerun with stratified randomization or keep an adjusted holdout because randomization checks show meaningful city imbalance among matched users.

Answers To The Practice Prompts

- An A/B test compares a control version against a treatment version with random user assignment.
- Workflow: define the hypothesis, choose the success metric, calculate sample size, randomize, run, analyze, then deploy or reject.
- A winning test needs p < 0.05 and a positive test-minus-control effect for the primary metric.
- Sample size depends on alpha, power, baseline metric, and minimum detectable effect; smaller effects need much larger samples.
- Randomization is checked by comparing distributions or by testing whether a model can predict treatment assignment.
- Novelty effect is a short-term lift from curiosity; check it by separating new and returning users.
- For multiple comparisons, use a stricter threshold such as Bonferroni alpha divided by the number of tests.
- Predefine hypotheses, metrics, sample size, and stopping rules; do not change them after seeing results.

Data Cleaning

Item	Count / treatment
Raw experiment rows	316,800
Treatment/price mismatch rows removed	365
Invalid timestamp rows flagged, retained for primary test	5,180
Clean rows used for primary conversion/revenue test	316,435
Rows without matching geography, retained as Unknown	41,141
Experiment date range	2015-03-02 to 2015-05-31

Main Results

Metric	Control	Test	Lift	Relative	p-value	95% CI
Conversion rate	1.99%	1.56%	-0.43%	-21.83%	< 1e-16	-0.53% to -0.34%
Revenue per visitor	$0.776	$0.918	$0.142	18.25%	1.10e-08	$0.093 to $0.190

Post-stratified by source, device, OS, and geo-match status, revenue per visitor increased from $0.776 to $0.923, a lift of $0.147 across 154 strata.

Randomization And Validity

Variable	Cramer's V	p-value	Interpretation
source	0.010	9.92e-04	small gap
device	0.016	< 1e-16	small gap
operative_system	0.031	< 1e-16	small gap
country	0.001	0.5517	small gap
geo_matched	0.001	0.5517	small gap
city_matched_only	0.090	< 1e-16	largest concern

The group differences in source, device, and OS are small in practical size, though statistically significant because the sample is large. City imbalance is more concerning and weakens a pure causal read of the raw averages.

Segment And Novelty Checks

The data does not include a new-vs-returning user flag, so the direct novelty-effect test recommended by the PDF cannot be performed. Weekly revenue lift is positive in many weeks but noisy, so a longer holdout or rerun is advisable.

Device Segments

Device	Control conv.	Test conv.	Control rev/user	Test rev/user	Rev. lift	p-value
mobile	1.99%	1.61%	$0.775	$0.953	$0.178	7.00e-08
web	1.99%	1.47%	$0.778	$0.870	$0.092	0.0142

Top Source Segments

Source	Control conv.	Test conv.	Control rev/user	Test rev/user	Rev. lift	p-value
direct_traffic	1.35%	1.01%	$0.527	$0.597	$0.070	0.1320
ads-google	2.26%	1.96%	$0.880	$1.158	$0.278	1.10e-05
ads_facebook	2.37%	1.69%	$0.923	$0.995	$0.072	0.2564
ads_other	1.54%	1.24%	$0.602	$0.731	$0.130	0.0729
seo-google	1.75%	1.60%	$0.682	$0.944	$0.262	0.0046
ads-bing	1.34%	0.96%	$0.521	$0.565	$0.044	0.5512
seo_facebook	1.75%	1.36%	$0.681	$0.802	$0.121	0.1743
friend_referral	4.17%	3.35%	$1.626	$1.975	$0.349	0.0124

Linux Sensitivity

Linux is 1.31% of the cleaned data and has 0% conversion in the test group, which the official solution treats as a likely segment-level issue. Removing Linux increases the test revenue per visitor from $0.918 to $0.934; the no-Linux revenue lift is $0.156, t-statistic 6.184, p-value 6.24e-10.

Sample	Control rev/user	Test rev/user	Rev. lift	t-stat	p-value
All cleaned rows	$0.776	$0.918	$0.142	5.715	1.10e-08
Linux removed	$0.778	$0.934	$0.156	6.184	6.24e-10

Bonus Stopping-Time Calculation

Using alpha 0.05, power 0.80, and the detectable conversion drop implied by moving from $39 to $59, the required sample is about 5,548 users per group. Average traffic is about 24356 users per week, so a two-week test needs about 11% of traffic in test. A minimum two-week runtime is still recommended to capture weekly patterns.

Final Recommendation

If revenue per visitor is the preselected North Star metric, the $59 price is economically and statistically attractive. The Linux sensitivity check strengthens the revenue result, but randomization is not perfectly balanced. The cleanest decision is to rerun the experiment with stratified assignment by city, device, OS, and traffic source, or to ship gradually with a permanent holdout and monitor conversion, revenue, referrals, refunds, and retention.